

Pandas Project

The Data

This data set contains booking information for a city hotel and a resort hotel, and includes information such as when the booking was made, length of stay, the number of adults, children, and/or babies, and the number of available parking spaces, among other things.

All personally identifying information has been removed from the data.

Source: Antonio, Nuno, Ana de Almeida, and Luis Nunes. "Hotel booking demand datasets." Data in brief 22 (2019): 41-49.

Data Column Reference

Variable	Type	Description	Source/Engineering
ADR	Numeric	Average Daily Rate as defined by [5]	BO, BL and TR / Calculated by dividing the sum of all lodging transactions by the total number of staying nights
Adults	Integer	Number of adults	BO and BL
Agent	Categorical	ID of the travel agency that made the booking ^a	BO and BL
ArrivalDateDayOfMonth	Integer	Day of the month of the arrival date	BO and BL
ArrivalDateMonth	Categorical	Month of arrival date with 12 categories: "January" to "December"	BO and BL
ArrivalDateWeekNumber	Integer	Week number of the arrival date	BO and BL
ArrivalDateYear	Integer	Year of arrival date	BO and BL
AssignedRoomType	Categorical	Code for the type of room assigned to the booking. Sometimes the assigned room type differs from the reserved room type due to hotel operation	BO and BL

Variable	Type	Description	Source/Engineering
		reasons (e.g. overbooking) or by customer request. Code is presented instead of designation for anonymity reasons	
Babies	Integer	Number of babies	BO and BL
BookingChanges	Integer	Number of changes/amendments made to the booking from the moment the booking was entered on the PMS until the moment of check-in or cancellation	BO and BL/Calculated by adding the number of unique iterations that change some of the booking attributes, namely: persons, arrival date, nights, reserved room type or meal
Children	Integer	Number of children	BO and BL/Sum of both payable and non-payable children
Company	Categorical	ID of the company/entity that made the booking or responsible for paying the booking. ID is presented instead of designation for anonymity reasons	BO and BL.
Country	Categorical	Country of origin. Categories are represented in the ISO 3155–3:2013 format [6]	BO, BL and NT
CustomerType	Categorical	Type of booking, assuming one of four categories: Contract - when the booking has an allotment or other type of contract associated to it; Group – when the booking is associated to a group;	BO and BL

Variable	Type	Description	Source/Engineering
		Transient – when the booking is not part of a group or contract, and is not associated to other transient booking;	
		Transient-party – when the booking is transient, but is associated to at least other transient booking	
<i>DaysInWaitingList</i>	Integer	Number of days the booking was in the waiting list before it was confirmed to the customer	BO/Calculated by subtracting the date the booking was confirmed to the customer from the date the booking entered on the PMS
<i>DepositType</i>	Categorical	Indication on if the customer made a deposit to guarantee the booking. This variable can assume three categories:	BO and TR/Value calculated based on the payments identified for the booking in the transaction (TR) table before the booking's arrival or cancellation date.
		No Deposit – no deposit was made;	In case no payments were found the value is "No Deposit".
		Non Refund – a deposit was made in the value of the total stay cost;	If the payment was equal or exceeded the total cost of stay, the value is set as "Non Refund".
		Refundable – a deposit was made with a value under the total cost of stay.	Otherwise the value is set as "Refundable"
<i>DistributionChannel</i>	Categorical	Booking distribution channel. The term "TA" means "Travel Agents" and "TO" means "Tour Operators"	BO, BL and DC
<i>IsCanceled</i>	Categorical	Value indicating if the booking was	BO

Variable	Type	Description	Source/Engineering
		canceled (1) or not (0)	
<i>IsRepeatedGuest</i>	Categorical	Value indicating if the booking name was from a repeated guest (1) or not (0)	BO, BL and C/ Variable created by verifying if a profile was associated with the booking customer. If so, and if the customer profile creation date was prior to the creation date for the booking on the PMS database it was assumed the booking was from a repeated guest
<i>LeadTime</i>	Integer	Number of days that elapsed between the entering date of the booking into the PMS and the arrival date	BO and BL/ Subtraction of the entering date from the arrival date
<i>MarketSegment</i>	Categorical	Market segment designation. In categories, the term "TA" means "Travel Agents" and "TO" means "Tour Operators"	BO, BL and MS
		Type of meal booked. Categories are presented in standard hospitality meal packages:	
		Undefined/SC – no meal package;	
<i>Meal</i>	Categorical	BB – Bed & Breakfast;	BO, BL and ML
		HB – Half board (breakfast and one other meal – usually dinner);	
		FB – Full board (breakfast, lunch and dinner)	

Variable	Type	Description	Source/Engineering
<i>PreviousBookingsNotCanceled</i>	Integer	Number of previous bookings not cancelled by the customer prior to the current booking	BO and BL / In case there was no customer profile associated with the booking, the value is set to 0. Otherwise, the value is the number of bookings with the same customer profile created before the current booking and not canceled.
<i>PreviousCancellations</i>	Integer	Number of previous bookings that were cancelled by the customer prior to the current booking	BO and BL/ In case there was no customer profile associated with the booking, the value is set to 0. Otherwise, the value is the number of bookings with the same customer profile created before the current booking and canceled.
<i>RequiredCardParkingSpaces</i>	Integer	Number of car parking spaces required by the customer	BO and BL
<i>ReservationStatus</i>	Categorical	Reservation last status, assuming one of three categories: Canceled – booking was canceled by the customer; Check-Out – customer has checked in but already departed; No-Show – customer did not check-in and did inform the hotel of the reason why	BO
<i>ReservationStatusDate</i>	Date	Date at which the last status was set. This variable can be used in conjunction with the <i>ReservationStatus</i> to understand when	BO

Variable	Type	Description	Source/Engineering
		was the booking canceled or when did the customer checked-out of the hotel	
ReservedRoomType	Categorical	Code of room type reserved. Code is presented instead of designation for anonymity reasons	BO and BL
StaysInWeekendNights	Integer	Number of weekend nights (Saturday or Sunday) the guest stayed or booked to stay at the hotel	BO and BL/ Calculated by counting the number of weekend nights from the total number of nights
StaysInWeekNights	Integer	Number of week nights (Monday to Friday) the guest stayed or booked to stay at the hotel	BO and BL/ Calculated by counting the number of week nights from the total number of nights
TotalOfSpecialRequests	Integer	Number of special requests made by the customer (e.g. twin bed or high floor)	BO and BL/Sum of all special requests

This Pandas project has twelve (12) questions. Answer all twelve.

Do not run the cells above the outputs!

1. Read the file "hotel_booking_data.csv" into a DataFrame. Show the first 4 rows of the DataFrame.

In [2]: `import pandas as pd`

```
df = pd.read_csv("CS767_HW4_PANDAS_hotel_booking_data.csv")
df.head(4)
```

Out [2]:

	hotel	is_canceled	lead_time	arrival_date_year	arrival_date_month	arrival_date_
0	Resort Hotel	0	342	2015	July	
1	Resort Hotel	0	737	2015	July	
2	Resort Hotel	0	7	2015	July	
3	Resort Hotel	0	13	2015	July	

4 rows × 36 columns

In [2]:

Out [2]:

	hotel	is_canceled	lead_time	arrival_date_year	arrival_date_month	arrival_date_
0	Resort Hotel	0	342	2015	July	
1	Resort Hotel	0	737	2015	July	
2	Resort Hotel	0	7	2015	July	
3	Resort Hotel	0	13	2015	July	

2. How many data points and features does this data set have? List the features of the data set.

In [3]: `print(f"This DataFrame has {df.shape[0]} rows and {df.shape[1]} columns")`

This DataFrame has 119390 rows and 36 columns

In [41]:

This DataFrame has 119390 rows and 36 columns

In [4]: `print(df.columns)`

```
Index(['hotel', 'is_canceled', 'lead_time', 'arrival_date_year',
      'arrival_date_month', 'arrival_date_week_number',
      'arrival_date_day_of_month', 'stays_in_weekend_nights',
      'stays_in_week_nights', 'adults', 'children', 'babies', 'meal',
      'country', 'market_segment', 'distribution_channel',
      'is_repeated_guest', 'previous_cancellations',
      'previous_bookings_not_canceled', 'reserved_room_type',
      'assigned_room_type', 'booking_changes', 'deposit_type', 'agent',
      'company', 'days_in_waiting_list', 'customer_type', 'adr',
      'required_car_parking_spaces', 'total_of_special_requests',
      'reservation_status', 'reservation_status_date', 'name', 'email',
      'phone-number', 'credit_card'],
      dtype='object')
```

In [42]:

```
Out[42]: Index(['hotel', 'is_canceled', 'lead_time', 'arrival_date_year',
      'arrival_date_month', 'arrival_date_week_number',
      'arrival_date_day_of_month', 'stays_in_weekend_nights',
      'stays_in_week_nights', 'adults', 'children', 'babies', 'meal',
      'country', 'market_segment', 'distribution_channel',
      'is_repeated_guest', 'previous_cancellations',
      'previous_bookings_not_canceled', 'reserved_room_type',
      'assigned_room_type', 'booking_changes', 'deposit_type', 'agent',
      'company', 'days_in_waiting_list', 'customer_type', 'adr',
      'required_car_parking_spaces', 'total_of_special_requests',
      'reservation_status', 'reservation_status_date', 'name', 'email',
      'phone-number', 'credit_card'],
      dtype='object')
```

3. Is there any missing data? If so, which column has the most missing data?

```
In [8]: print(df.isnull().sum().idxmax())
```

company

In [12]:

```
Out[12]: 'company'
```

4. Print the non-null count and data type of each column.

```
In [9]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 119390 entries, 0 to 119389
Data columns (total 36 columns):
 #   Column                                  Non-Null Count  Dtype
---  -
 0   hotel                                  119390 non-null  object
 1   is_canceled                           119390 non-null  int64
 2   lead_time                             119390 non-null  int64
 3   arrival_date_year                     119390 non-null  int64
 4   arrival_date_month                    119390 non-null  object
 5   arrival_date_week_number              119390 non-null  int64
 6   arrival_date_day_of_month             119390 non-null  int64
 7   stays_in_weekend_nights               119390 non-null  int64
 8   stays_in_week_nights                  119390 non-null  int64
 9   adults                                119390 non-null  int64
10  children                              119386 non-null  float64
11  babies                                119390 non-null  int64
12  meal                                  119390 non-null  object
13  country                              118902 non-null  object
14  market_segment                        119390 non-null  object
15  distribution_channel                  119390 non-null  object
16  is_repeated_guest                     119390 non-null  int64
17  previous_cancellations                 119390 non-null  int64
18  previous_bookings_not_canceled         119390 non-null  int64
19  reserved_room_type                    119390 non-null  object
20  assigned_room_type                    119390 non-null  object
21  booking_changes                       119390 non-null  int64
22  deposit_type                           119390 non-null  object
23  agent                                 103050 non-null  float64
24  company                               6797 non-null   float64
25  days_in_waiting_list                   119390 non-null  int64
26  customer_type                          119390 non-null  object
27  adr                                    119390 non-null  float64
28  required_car_parking_spaces            119390 non-null  int64
29  total_of_special_requests              119390 non-null  int64
30  reservation_status                    119390 non-null  object
31  reservation_status_date                119390 non-null  object
32  name                                   119390 non-null  object
33  email                                  119390 non-null  object
34  phone_number                           119390 non-null  object
35  credit_card                            119390 non-null  object
dtypes: float64(4), int64(16), object(16)
memory usage: 32.8+ MB

```

In [9]:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 119390 entries, 0 to 119389
Data columns (total 36 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   hotel                                119390 non-null  object
1   is_canceled                          119390 non-null  int64
2   lead_time                           119390 non-null  int64
3   arrival_date_year                    119390 non-null  int64
4   arrival_date_month                   119390 non-null  object
5   arrival_date_week_number             119390 non-null  int64
6   arrival_date_day_of_month            119390 non-null  int64
7   stays_in_weekend_nights              119390 non-null  int64
8   stays_in_week_nights                 119390 non-null  int64
9   adults                               119390 non-null  int64
10  children                             119386 non-null  float64
11  babies                               119390 non-null  int64
12  meal                                 119390 non-null  object
13  country                              118902 non-null  object
14  market_segment                       119390 non-null  object
15  distribution_channel                 119390 non-null  object
16  is_repeated_guest                    119390 non-null  int64
17  previous_cancellations                119390 non-null  int64
18  previous_bookings_not_canceled        119390 non-null  int64
19  reserved_room_type                   119390 non-null  object
20  assigned_room_type                   119390 non-null  object
21  booking_changes                       119390 non-null  int64
22  deposit_type                         119390 non-null  object
23  agent                                103050 non-null  float64
24  company                              6797 non-null   float64
25  days_in_waiting_list                 119390 non-null  int64
26  customer_type                        119390 non-null  object
27  adr                                   119390 non-null  float64
28  required_car_parking_spaces          119390 non-null  int64
29  total_of_special_requests            119390 non-null  int64
30  reservation_status                  119390 non-null  object
31  reservation_status_date              119390 non-null  object
32  name                                 119390 non-null  object
33  email                                119390 non-null  object
34  phone-number                         119390 non-null  object
35  credit_card                          119390 non-null  object
dtypes: float64(4), int64(16), object(16)
memory usage: 32.8+ MB
```

5. Remove the 'company' column from the data set.

```
In [10]: df = df.drop(columns=['company'])
```

6. What are the top 5 most common country codes in the dataset? Answer using a single line of code.

```
In [11]: df['country'].value_counts().head(5)
```

```
Out[11]: country
PRT      48590
GBR      12129
FRA      10415
ESP       8568
DEU       7287
Name: count, dtype: int64
```

```
In [19]:
```

```
Out[19]: country
PRT      48590
GBR      12129
FRA      10415
ESP       8568
DEU       7287
Name: count, dtype: int64
```

7. The adr is the average daily rate for a person's stay at the hotel. What is the mean adr across all the hotel stays in the dataset? Round this to 2 decimal points.

```
In [12]: print(round(df['adr'].mean(), 2))
```

```
101.83
```

```
In [43]:
```

```
Out[43]: 101.83
```

8. What is the average (mean) number of nights for a stay across the entire data set? Round this to 2 decimal points.

```
In [13]: print(round((df['stays_in_weekend_nights'] + df['stays_in_week_nights']).mean(), 2))
```

```
3.43
```

```
In [27]:
```

```
Out[27]: 3.43
```

9. What is the average *total* cost for a stay in the dataset? Not average daily cost, but *total stay* cost. (You will need to calculate total cost your self by using ADR and week day and weeknight stays). Round this to 2 decimal points.

```
In [14]: print(round((df['adr'] * (df['stays_in_weekend_nights'] + df['stays_in_week_nights'])).mean(), 2))
```

```
357.85
```

```
In [25]:
```

```
357.85
```

10. What are the names and emails of people who made exactly 5 "Special Requests"?

```
In [15]: df[df['total_of_special_requests'] == 5][['name', 'email']]
```

Out [15]:

	name	email
7860	Amanda Harper	Amanda.H66@yahoo.com
11125	Laura Sanders	Sanders_Laura@hotmail.com
14596	Tommy Ortiz	Tommy_O@hotmail.com
14921	Gilbert Miller	Miller.Gilbert@aol.com
14922	Timothy Torres	TTorres@protonmail.com
24630	Jennifer Weaver	Jennifer_W@aol.com
27288	Crystal Horton	Crystal.H@mail.com
27477	Brittney Burke	Burke_Brittney16@att.com
29906	Cynthia Cabrera	Cabrera.Cynthia@xfinity.com
29949	Sarah Floyd	Sarah_F@gmail.com
32267	Michelle Villa	Michelle.Villa@aol.com
39027	Nichole Hebert	Hebert.Nichole@gmail.com
39129	Lindsey Mckenzie	Lindsey.Mckenzie@att.com
39525	Ashley Edwards	Edwards.Ashley@yahoo.com
70114	Christopher Torres	Torres.Christopher@gmail.com
78819	Mrs. Tara Sullivan DVM	Mrs..DVM@xfinity.com
78820	Michaela Brown	MichaelaBrown@att.com
78822	Kurt Maldonado MD	KMD15@xfinity.com
97072	Jason Richardson	Jason.R@zoho.com
97099	Terri Hurley	THurley@xfinity.com
97261	Mrs. Caitlin Webb	Mrs._W@comcast.net
98410	Holly Arroyo	Arroyo_Holly@mail.com
98674	Denise Campbell	Denise_C@gmail.com
99887	Michael Smith	Michael.S42@aol.com
99888	Dr. Trevor Sellers	Dr._S@aol.com
101569	Kayla Murphy	Kayla.Murphy@yahoo.com
102061	Taylor Martinez	Taylor.Martinez@hotmail.com
109511	Charles Wilson	Charles_Wilson@yahoo.com
109590	Tyler Allison	Tyler.A@protonmail.com
110082	Matthew Bailey	Matthew_Bailey@aol.com
110083	Charlotte Acevedo	Charlotte_A@verizon.com
111909	Darrell Brennan	Brennan_Darrell51@hotmail.com

	name	email
111911	Melinda Jensen	MelindaJensen@zoho.com
113915	Terry Arnold	Arnold.Terry@zoho.com
114770	Mary Nguyen	Nguyen.Mary@protonmail.com
114909	Lindsay Cuevas	Lindsay.Cuevas40@mail.com
116455	Cynthia Hernandez	CynthiaHernandez@xfinity.com
116457	Angela Hawkins	Angela_H@gmail.com
118817	Sue Lawson	Sue.L52@comcast.net
119161	Alyssa Richards	Alyssa_Richards@aol.com

In [29]:

Out [29] :

	name	email
7860	Amanda Harper	Amanda.H66@yahoo.com
11125	Laura Sanders	Sanders_Laura@hotmail.com
14596	Tommy Ortiz	Tommy_O@hotmail.com
14921	Gilbert Miller	Miller.Gilbert@aol.com
14922	Timothy Torres	TTorres@protonmail.com
24630	Jennifer Weaver	Jennifer_W@aol.com
27288	Crystal Horton	Crystal.H@mail.com
27477	Brittney Burke	Burke_Brittney16@att.com
29906	Cynthia Cabrera	Cabrera.Cynthia@xfinity.com
29949	Sarah Floyd	Sarah_F@gmail.com
32267	Michelle Villa	Michelle.Villa@aol.com
39027	Nichole Hebert	Hebert.Nichole@gmail.com
39129	Lindsey Mckenzie	Lindsey.Mckenzie@att.com
39525	Ashley Edwards	Edwards.Ashley@yahoo.com
70114	Christopher Torres	Torres.Christopher@gmail.com
78819	Mrs. Tara Sullivan DVM	Mrs..DVM@xfinity.com
78820	Michaela Brown	MichaelaBrown@att.com
78822	Kurt Maldonado MD	KMD15@xfinity.com
97072	Jason Richardson	Jason.R@zoho.com
97099	Terri Hurley	THurley@xfinity.com
97261	Mrs. Caitlin Webb	Mrs._W@comcast.net
98410	Holly Arroyo	Arroyo_Holly@mail.com
98674	Denise Campbell	Denise_C@gmail.com
99887	Michael Smith	Michael.S42@aol.com
99888	Dr. Trevor Sellers	Dr._S@aol.com
101569	Kayla Murphy	Kayla.Murphy@yahoo.com
102061	Taylor Martinez	Taylor.Martinez@hotmail.com
109511	Charles Wilson	Charles_Wilson@yahoo.com
109590	Tyler Allison	Tyler.A@protonmail.com
110082	Matthew Bailey	Matthew_Bailey@aol.com
110083	Charlotte Acevedo	Charlotte_A@verizon.com
111909	Darrell Brennan	Brennan_Darrell51@hotmail.com

	name	email
111911	Melinda Jensen	MelindaJensen@zoho.com
113915	Terry Arnold	Arnold.Terry@zoho.com
114770	Mary Nguyen	Nguyen.Mary@protonmail.com
114909	Lindsay Cuevas	Lindsay.Cuevas40@mail.com
116455	Cynthia Hernandez	CynthiaHernandez@xfinity.com
116457	Angela Hawkins	Angela_H@gmail.com
118817	Sue Lawson	Sue.L52@comcast.net
119161	Alyssa Richards	Alyssa_Richards@aol.com

11. What are the top 5 most common last name in the dataset? Bonus: Can you figure this out in one line of pandas code?

- For simplicity treat the a title such as MD as a last name, for example Caroline Conley MD can be said to have the last name MD.
- You may have to revisit string methods and lambda functions for this question.

```
In [16]: df['name'].apply(lambda x: x.split()[-1]).value_counts().head(5)
```

```
Out[16]: name
Smith      2503
Johnson   1990
Williams   1618
Jones      1434
Brown      1423
Name: count, dtype: int64
```

```
In [34]:
```

```
Out[34]: name
Smith      2503
Johnson   1990
Williams   1618
Jones      1434
Brown      1423
Name: count, dtype: int64
```

12. How many arrivals took place between the 1st and the 15th of the month (inclusive of 1 and 15)? Bonus: Can you do this in one line of pandas code?

```
In [17]: df[(df['arrival_date_day_of_month'] >= 1) & (df['arrival_date_day_of_month']
```

```
Out[17]: 58152
```

```
In [37]:
```

Out[37]: 58152